

Practical 1: To analyze the dataset by performing exploratory data analysis (EDA)

Steps for EDA:

Data Collection and Cleaning: The first step in EDA is to collect the data and clean it. Data cleaning involves removing missing or erroneous data, handling outliers, and converting data types to make it ready for analysis.

Data Exploration: Once the data has been cleaned, the next step is to explore it. Data exploration involves understanding the data and its features, such as understanding the distribution of data, identifying patterns, and checking for correlations between variables.

Visualization: Visualizations are an integral part of EDA. Visualization tools help users to understand the data better. For instance, users can create scatter plots, histograms, box plots, heatmaps, and more, to explore the data visually.

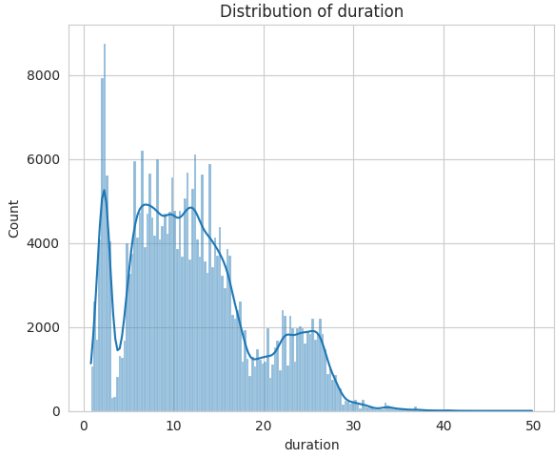
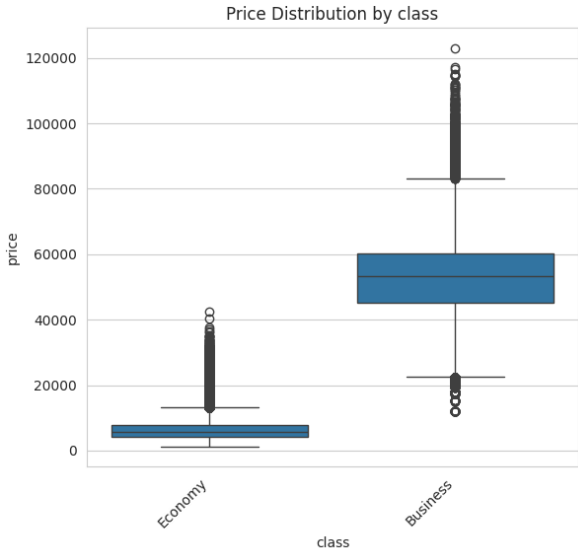
Feature Engineering: Feature engineering is an essential part of EDA, which involves creating new features based on the existing data. Feature engineering can help improve the performance of models by providing them with more relevant data.

Hypothesis Testing: Hypothesis testing is the process of testing a hypothesis or claim about a population parameter. It can help to confirm or refute assumptions made during the EDA process.

Conclusion: Finally, the last step in EDA is to draw conclusions based on the insights gained from the analysis. These conclusions can be used to make decisions or to further investigate the data.

No of rows: 500

No of features/parameters: 11

Visualization Chart		Inference	
1. UNIVARIATE GRAPH/S			
Histograms 		The distribution of flight durations is right-skewed, with most flights lasting between 1–12 hours, peaking sharply around 2–4 hours for short-haul routes, and a smaller cluster at 20–28 hours likely representing long-haul or multi-leg journeys. Very long flights beyond 30 hours are rare, indicating that the airline network is dominated by short- to medium-duration flights with limited ultra-long routes.	
Box Plot 		The boxplot shows that Business class ticket prices are significantly higher than Economy, with a median around ₹55,000 compared to about ₹5,000 for Economy. Both classes have many outliers, but Business class has a wider price range and more extreme high values, while Economy prices are more concentrated at the lower end.	

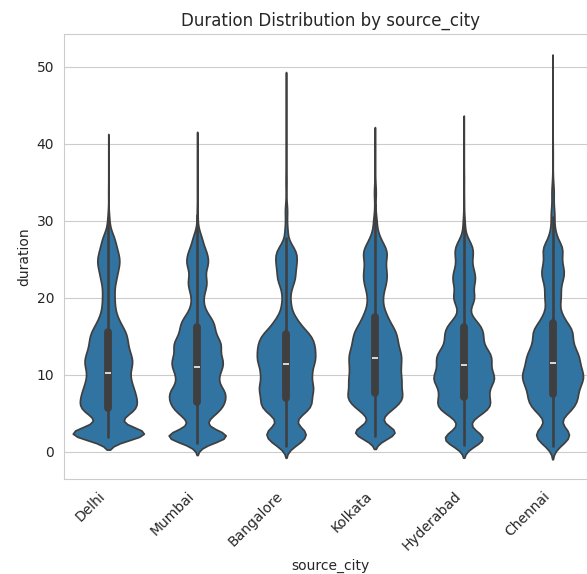
2. BIVARIATE GRAPH/S

Scatter Plot



The scatter plot shows the relationship between flight duration and price. The majority of flights are clustered between 5–25 hours in duration, with prices spread widely from low to very high. Very short and very long flights are fewer, and extreme prices are visible as outliers.

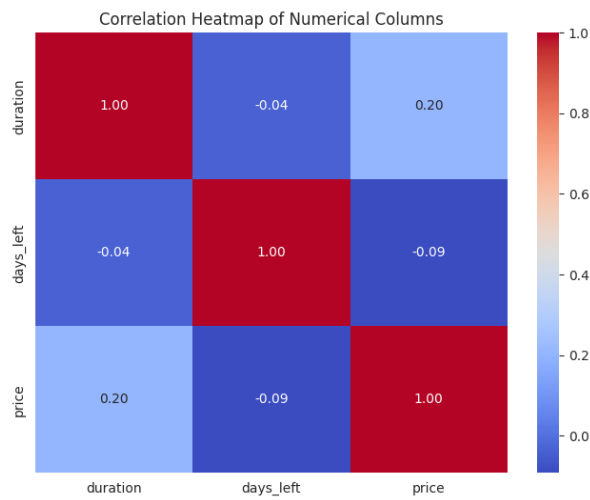
Violin Plot



The violin chart compares flight durations across different source cities, where the width at each point shows how frequently a duration occurs, the white dot marks the median, the thick bar shows the middle 50% of durations, and the thin line spans the full range. Wider sections mean more flights of that duration, and differences in shapes reveal variation in duration patterns between cities.

3. MULTIVARIATE GRAPH/S

Heatmap (Correlation matrix)



Values range from -1 to 1, where positive numbers indicate a direct relationship and negative numbers indicate an inverse relationship. Here, flight duration and price have a weak positive correlation (0.20), meaning longer flights tend to cost slightly more.